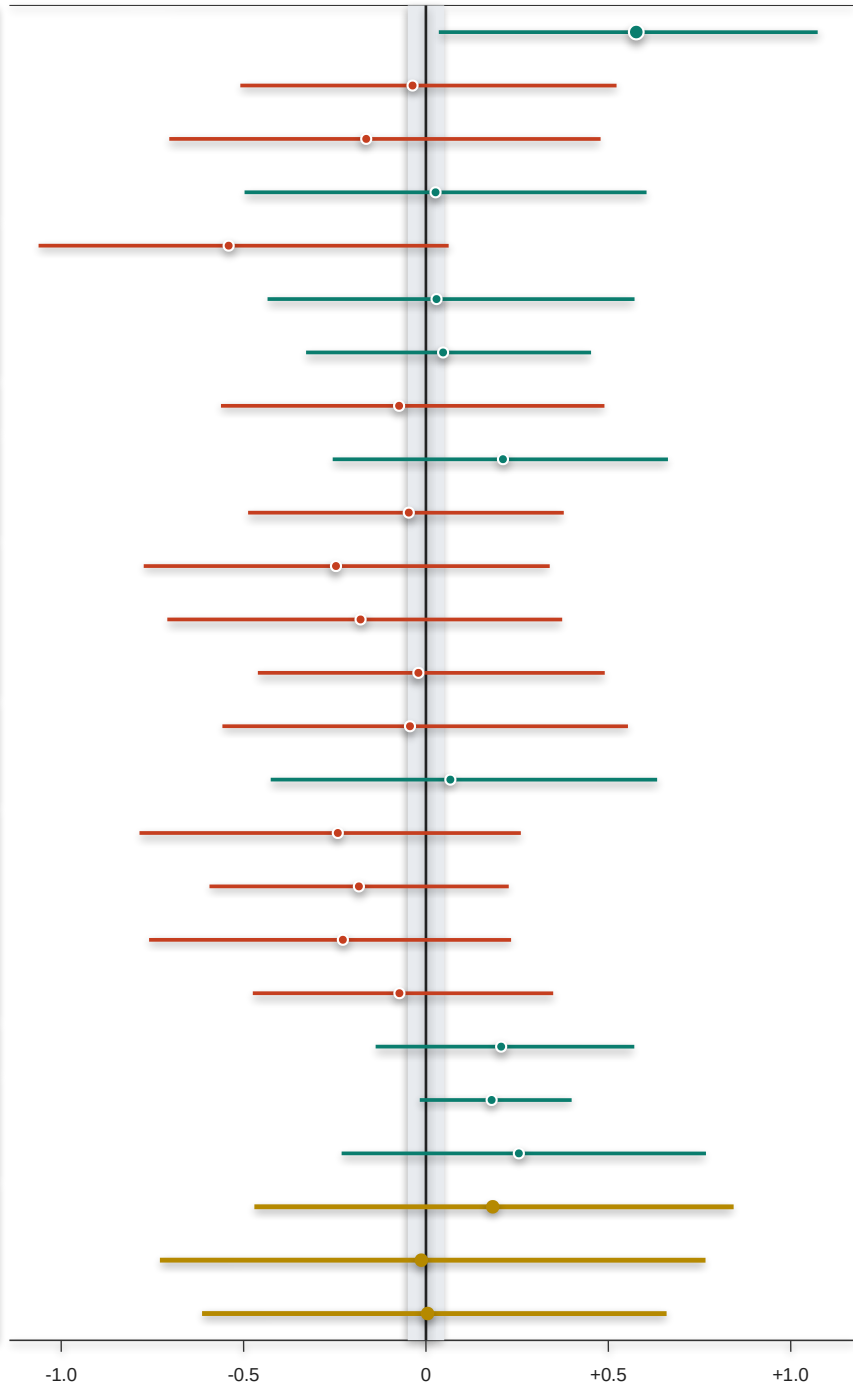


|                      |                         |
|----------------------|-------------------------|
| Cytotoxic T-cell     | CD8 proliferation       |
|                      | CD8 exhaustion          |
|                      | Cytolytic activity      |
|                      | CD8 activation          |
| Antigen presentation | MHC-II                  |
|                      | NLRC5 · HLA · IFNG      |
|                      | Antigen presentation    |
| IFN response         | IFN-γ (Ayers 18)        |
| B-cell / TLS         | B cell                  |
|                      | TLS (Cabrita)           |
| Innate               | Macrophage M2           |
|                      | NK cell                 |
|                      | Macrophage M1           |
| Regulatory           | Checkpoint inhibitory   |
|                      | Treg                    |
| Stromal/EMT          | CAF — myCAF             |
|                      | EMT (Mak)               |
|                      | TGF-β (Mariathasan)     |
|                      | CAF — iCAF              |
| Hypoxia              | Hypoxia (Buffa)         |
| Other                | Stemness (mRNAsi proxy) |
|                      | Epithelial              |
| Purified immune      | B-cell infiltration ★   |
|                      | CD8 cytotoxic ★         |
|                      | T-cell infiltration ★   |



| Δ     | 95 % CI        | U   | P (MW)   | q (BH) |
|-------|----------------|-----|----------|--------|
| +0.58 | [+0.04, +1.07] | 195 | 0.0351 ★ | 0.814  |
| -0.04 | [-0.51, +0.52] | 106 | 0.288    | 0.814  |
| -0.16 | [-0.70, +0.48] | 107 | 0.305    | 0.814  |
| +0.03 | [-0.50, +0.60] | 141 | 0.871    | 0.957  |
| -0.54 | [-1.06, +0.06] | 86  | 0.0746   | 0.814  |
| +0.03 | [-0.43, +0.57] | 131 | 0.871    | 0.957  |
| +0.05 | [-0.33, +0.45] | 133 | 0.928    | 0.957  |
| -0.07 | [-0.56, +0.49] | 122 | 0.627    | 0.906  |
| +0.21 | [-0.26, +0.66] | 175 | 0.165    | 0.814  |
| -0.05 | [-0.49, +0.38] | 138 | 0.957    | 0.957  |
| -0.25 | [-0.77, +0.34] | 100 | 0.201    | 0.814  |
| -0.18 | [-0.71, +0.37] | 110 | 0.358    | 0.814  |
| -0.02 | [-0.46, +0.49] | 123 | 0.653    | 0.906  |
| -0.04 | [-0.56, +0.55] | 123 | 0.653    | 0.906  |
| +0.07 | [-0.43, +0.63] | 123 | 0.653    | 0.906  |
| -0.24 | [-0.79, +0.26] | 103 | 0.242    | 0.814  |
| -0.18 | [-0.59, +0.23] | 110 | 0.358    | 0.814  |
| -0.23 | [-0.76, +0.23] | 116 | 0.482    | 0.902  |
| -0.07 | [-0.47, +0.35] | 128 | 0.787    | 0.957  |
| +0.21 | [-0.14, +0.57] | 166 | 0.288    | 0.814  |
| +0.18 | [-0.02, +0.40] | 177 | 0.145    | 0.814  |
| +0.25 | [-0.23, +0.77] | 159 | 0.418    | 0.87   |
| +0.18 | [-0.47, +0.84] | 155 | 0.505    | 0.902  |
| -0.01 | [-0.73, +0.77] | 142 | 0.843    | 0.957  |
| +0.00 | [-0.61, +0.66] | 134 | 0.957    | 0.957  |

★ purified immune signatures — CD8 cytotoxic / T-cell infiltration / B-cell infiltration, externally validated in §3.12

Δ > 0 (good-enriched) Δ < 0 (poor-enriched) ±0.05 no-effect zone

Stars: ★  $P < 0.05$  · ★★  $P < 0.01$  · ★★★  $P < 0.001$  ·  $q$  = Benjamini–Hochberg FDR across 25 signatures ·  $\Delta$  = mean(good) – mean(poor) on per-sample z-score; 1,000×